Advanced Computer-Aided VLSI System Design
Lab5: Innovus Lab (2/2)

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Data Preparation

1. Extract data from the folder **Innovus_Lab**.
2. The extracted directory contains
   - **design**
     A. CHIP_syn.v
     B. CHIP.io
     C. CHIP_func.sdc
     D. CHIP_scan.sdc
     E. CHIP-scan.def
   - **library**
     - **lef**
       A. N16ADFP_APR_Innovus_11M.10a.tlef
       B. N16ADFP_BondPad.lef
       C. N16ADFP_StdCell.lef
       D. N16ADFP_StdIO.lef
     - **lib**
       A. N16ADFP_StdCellff0p88v125c_ccs.lib
       B. N16ADFP_StdCellff0p88vm40c_ccs.lib
       C. N16ADFP_StdCellss0p72v125c_ccs.lib
       D. N16ADFP_StdCellss0p72vm40c_ccs.lib
       E. N16ADFP_StdIOff0p88v1p98v125c.lib
       F. N16ADFP_StdIOff0p88v1p98vm40c.lib
       G. N16ADFP_StdIOss0p72v1p62v125c.lib
       H. N16ADFP_StdIOss0p72v1p62vm40c.lib
     - **memory**
       A. rf_2p_hse.lef
       B. rf_2p_hse_nldm_ff_0p88v_0p88v_125c_syn.lib
       C. rf_2p_hse_nldm_ff_0p88v_0p88v_m40c_syn.lib
       D. rf_2p_hse_nldm_ss_0p72v_0p72v_125c_syn.lib
       E. rf_2p_hse_nldm_ss_0p72v_0p72v_m40c_syn.lib
       F. rf_2p_hse.gds
   - **sim**
     - **pre_sim**
       A. CHIP.tcf

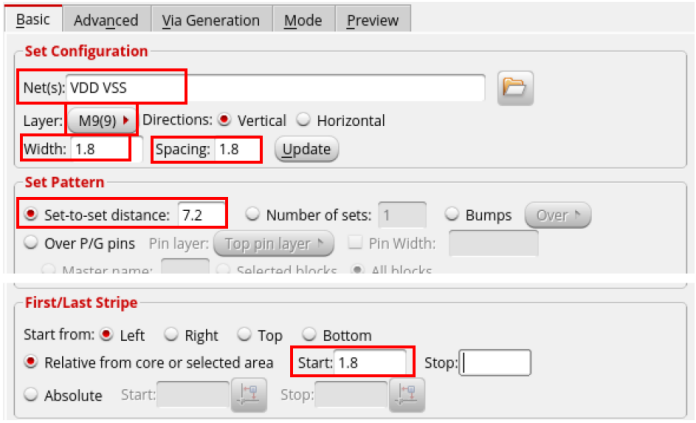
- **post_sim**
  A. a.map
  B. CHIP_sim.v
  C. golden_result
  D. parameter.v
  E. vlog.f
  F. xrun.tcl
- mmmc.view

Introduction

In this lab, you will learn how to use Innovus to run the APR flow in N16 ADFP, and generate the required data for demonstration.

Power Planning: Stripe

1. Add stripes from M9 to M5. Open ***Power → Power Planning → Add Stripes…***
   - Net(s): **VDD VSS**
   - Layer: **M9(9)**
   - Width: **1.8** Spacing: **1.8**
   - Set-to-set distance: **7.2**
   - Start: **1.8**

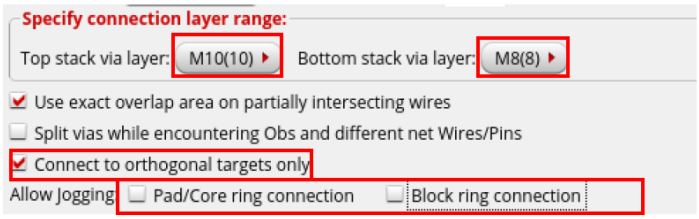


   - Choose ***Advanced*** in the window
   - Choose ◆ **Merge with block rings if spacing less than: 10**
   - Choose ◆ **Auto**

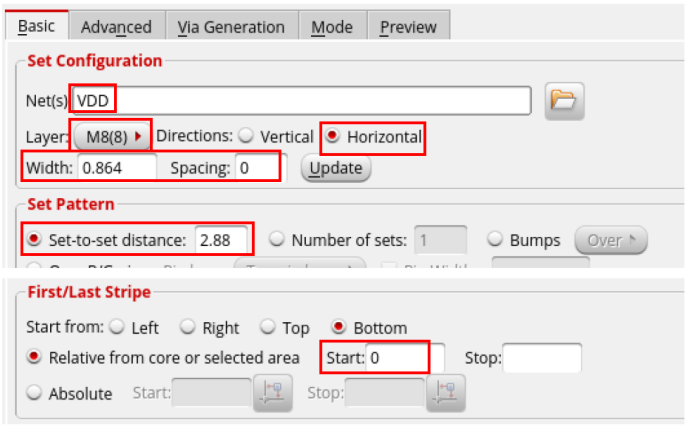


   - Choose ***Mode*** in the window
   - Top stack via layer: **M10(10)** Bottom stack via layer: **M8(8)**
   - Choose ◆ **Connect to orthogonal targets only**

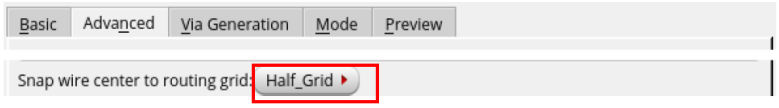
- Unchosen **Pad/Core ring connection** and **Block ring connection**
- Click Apply button



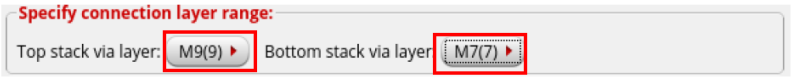
- Choose ***Basic*** in the window and set as following figure to add horizontal M8 VDD stripe.



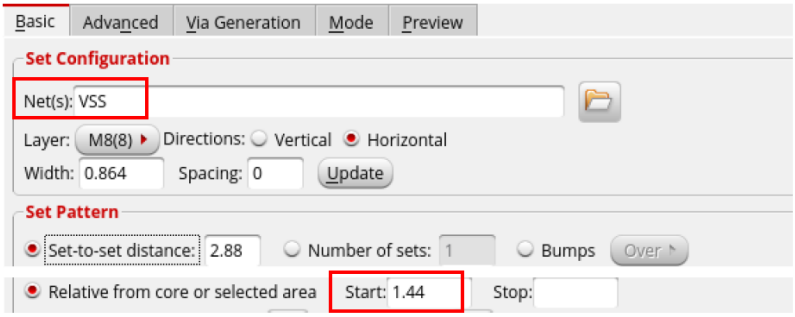
- Choose ***Advanced*** in the window
- Snap wire center to routing grid: **Half_Grid**



- Choose ***Mode*** in the window
- Top stack via layer: **M9(9)** Bottom stack via layer: **M7(7)**
- Click Apply button



- Choose ***Basic*** in the window and set as following figure to add horizontal M8 VSS stripe.
- Click Apply button



- Follow the table to add M7 M6 M5 stripes. (M9 and M8 stripes have been added.)

Net	Metal layer	dir	Start	Width	Space	Set To Set	Top stack via	Bottom stack via	Snap to routing grid
VDD VSS	M9	V	1.8	1.8	1.8	7.2	M10	M8	
VDD	M8	H	0	0.864	0	2.88	M9	M7	Half_Grid
VSS	M8	H	1.44	0.864	0	2.88	M9	M7	Half_Grid
VDD	M7	V	0	0.24	0	7.2	M8	M6	Grid
VSS	M7	V	3.6	0.24	0	7.2	M8	M6	Grid
VDD	M6	H	0	0.24	0	7.2	M7	M5	Grid
VSS	M6	H	3.6	0.24	0	7.2	M7	M5	Grid
VDD	M5	V	0	0.24	0	7.2	M6	M4	Grid
VSS	M5	V	3.6	0.24	0	7.2	M6	M4	Grid

- this

2. Save file
   - Open ***File → Save Design…***
   - Choose ◆ Innovus
   - File Name: **DBS/powerplan.enc**

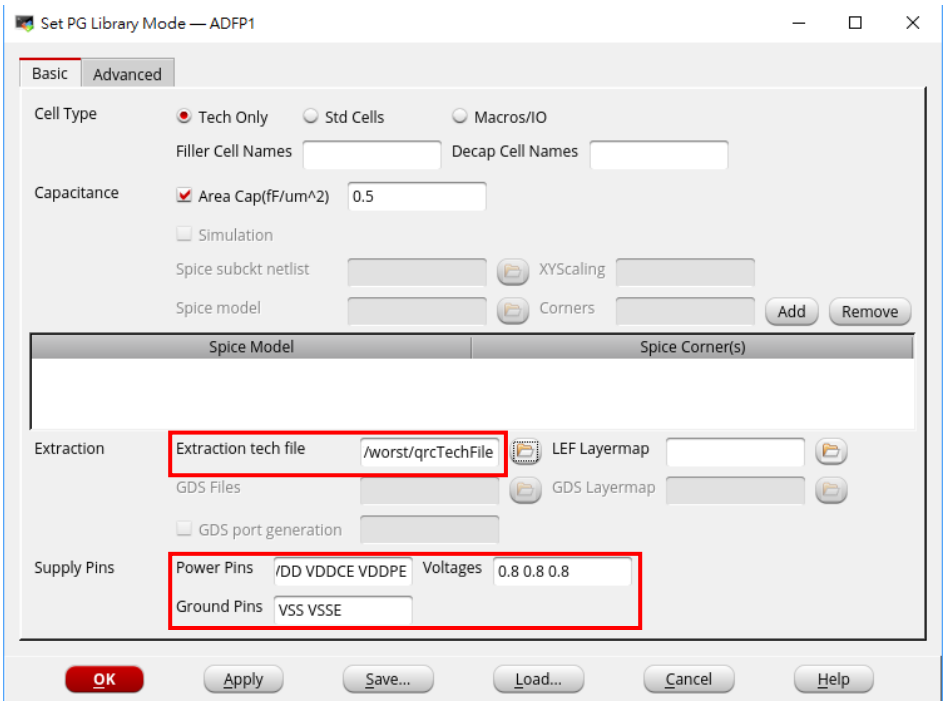
Early Rail Analysis

1. Power analysis
   - innovus # > **source lab_script/set_activity.tcl**
   - innovus # > **report_power**
2. Find activity annotation summary in the terminal and the power analysis result. The higher annotation ratio means it is closer to simulation situation. You should see **static_VDD.ptiavg** and **static_VSS.ptiavg**, which contain the power consumption information.

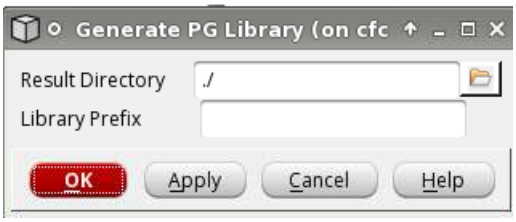
```
Activity annotation summary:
       Primary Inputs : 7/8 = 87.5%
         Flop outputs : 6584/6585 = 99.9848%
 Memory/Macro outputs : 352/352 = 100%
     Tristate outputs : 0/0 = 0%
           Total Nets : 20557/22250 = 92.391%

Total Internal Power:        7.81832897            65.8426%
Total Switching Power:       0.29024370             2.4443%
Total Leakage Power:         3.76570340            31.7131%
Total Power:                11.87427606
```

3. Open ***Power → Rail Analysis → Set PG Library Mode***
   - Extraction tech file: **/share1/tech/ADFP/Executable_Package/Collaterals/Tech/RC/N16ADFP_QRC/worst/qrcTechFile**
   - Power Pins: **VDD VDDCE VDDPE** Voltage: **0.8 0.8 0.8**
   - Ground Pins: **VSS VSSE**
   - Click OK button



4. Open ***Power → Rail Analysis → Generate PG Library***
   - ➢ Click OK button



   Note: **techonly.cl** is generated after this step.

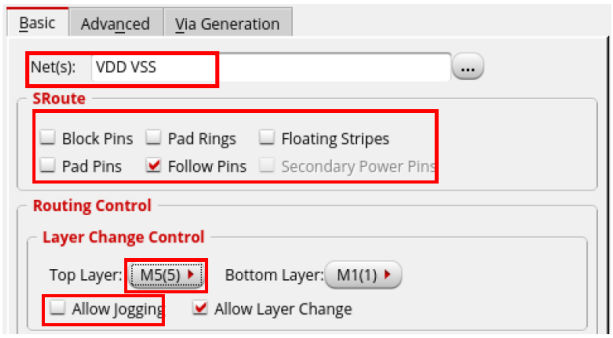
5. innovus # > **source lab_script/early_rail_analysis.tcl**
6. Open **PD_25C_avg_1/Reports/report_generation.log** to find IR Drop result. If IR drop is OK, then move on to next step.

```
Net VSS:
      Voltage: 0
      Threshold: 0.04
      Minimum, Average, Maximum IR Drop: 0.000V, 0.829mV, 1.544mV
        Range 1(    1.351mV  -      1.544mV ):      1671 (  2.64%)
        Range 2(    1.158mV  -      1.351mV ):     11080 ( 17.52%)
        Range 3(    0.965mV  -      1.158mV ):     12203 ( 19.29%)
        Range 4(    0.772mV  -      0.965mV ):     10535 ( 16.66%)
        Range 5(    0.579mV  -      0.772mV ):     11205 ( 17.72%)
        Range 6(    0.386mV  -      0.579mV ):     10646 ( 16.83%)
        Range 7(    0.193mV  -      0.386mV ):      4263 (  6.74%)
        Range 8(     0.000V  -      0.193mV ):      1647 (  2.60%)

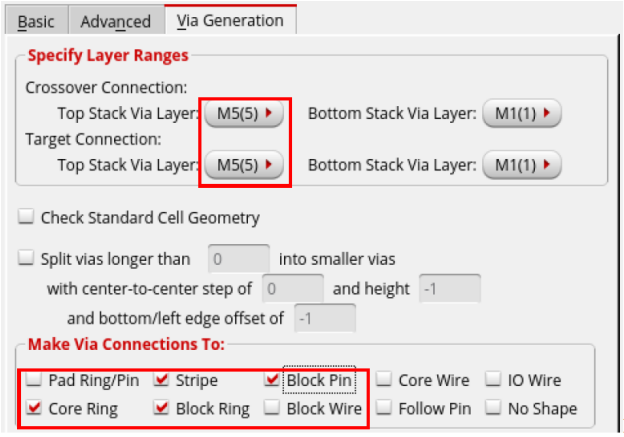
Net VDD:
      Voltage: 0.8
      Threshold: 0.76
      Minimum, Average, Maximum IR Drop: 0.799V, 0.799V, 0.800V
        Range 1(     0.799V  -       0.799V ):      3578 (  5.07%)
        Range 2(     0.799V  -       0.799V ):     12365 ( 17.51%)
        Range 3(     0.799V  -       0.799V ):     14230 ( 20.15%)
        Range 4(     0.799V  -       0.799V ):     10867 ( 15.39%)
        Range 5(     0.799V  -       0.799V ):     12752 ( 18.06%)
        Range 6(     0.799V  -       0.800V ):      9689 ( 13.72%)
        Range 7(     0.800V  -       0.800V ):      4722 (  6.69%)
        Range 8(     0.800V  -       0.800V ):      2407 (  3.41%)
```

Power Planning: Follow Pin

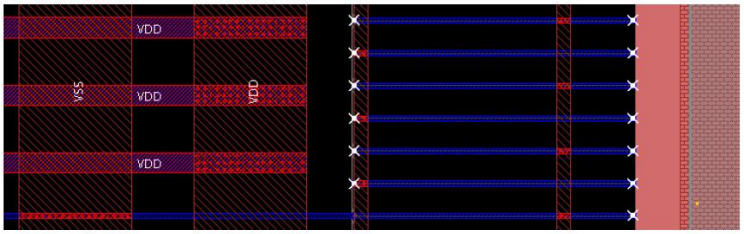
1. Add M1 follow pin. Open ***Route → Special Route…***
   - Net(s): **VDD VSS**
   - Preserve only ◆ **Follow Pins**
   - Top Layer: **M5(5)**
   - Unchosen **Allow jogging**



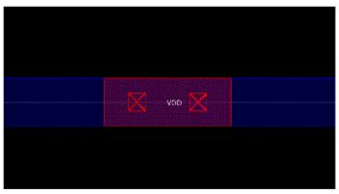
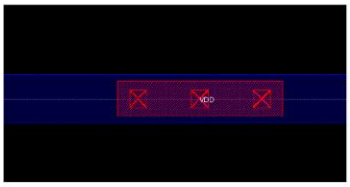
   - Choose ***Via Generation*** in the window
   - Top Stack Via Layer: **M5(5)**
   - Choose ◆ **Stripe** ◆ **Block Pin** ◆ **Core Ring** ◆ **Block Ring**
   - Click OK button



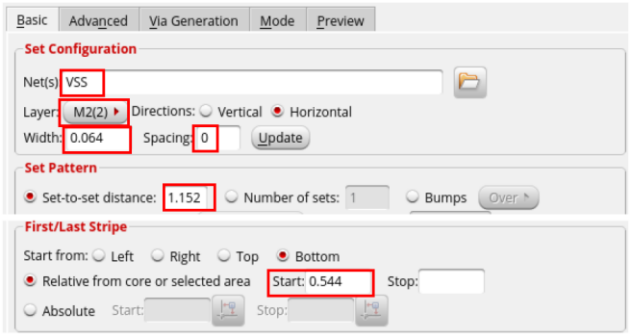
2. In the area where the pad pin connects to the power ring and the boundary of the SRAM, there are some "X" indicating that these follow pins are not connected at the ends. These "X" can be ignored. Click **Shift+V** to waive.



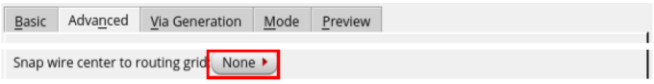
3. Shrink VIA1 to preserve more space for routing. (Can turn on the visibility of M1, VIA12 and M2 and turn off the other metals visibility to observe the change.)
   - innovus # > **source lab_script/shrink_via.tcl**

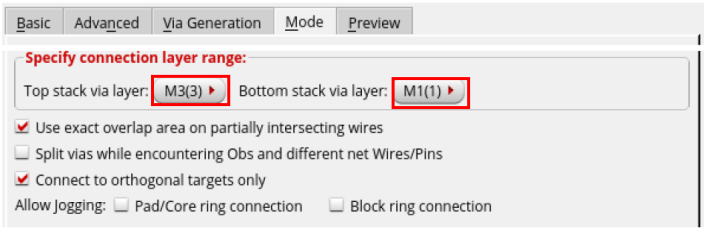
4. Add M2 follow pin. Open ***Power → Power Planning → Add Stripes…***
   - Set as the following figure



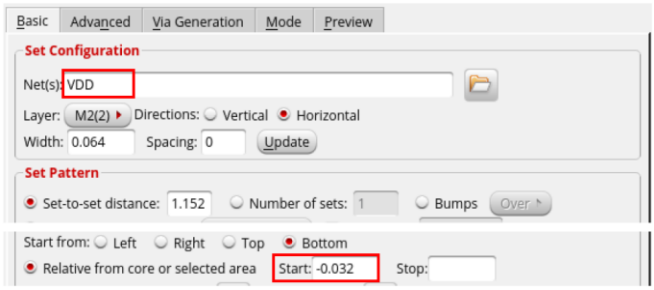
   - Choose ***Advanced*** in the window
   - Snap wire center to routing grid: **None**



   - Choose ***Mode*** in the window
   - Top stack via layer: **M3(3)** Bottom stack via layer: **M1(1)**
   - Click Apply button



   Note: M2 stripe must completely on M1 followpin.
   - Choose ***Basic*** in the window
   - Net(s): **VDD**
   - Start: **-0.032**
   - Click OK button



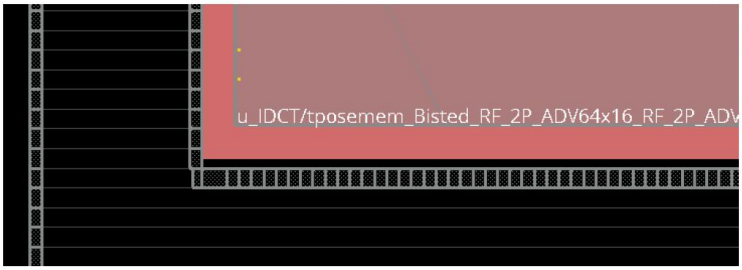
5. Delete routing blockage
   - innovus # > **deleteRouteBlk -name RBK***

6. Open ***Verify → Verify DRC…***
   - Click OK button. Fix all DRC errors
7. Open ***File → Save → Floorplan…***
   - File name: **powerplan.fp**
   - Click Save button
8. Save file
   - Open ***File → Save Design…***
   - Choose ◆ Innovus
   - File Name: **DBS/followpin.enc**

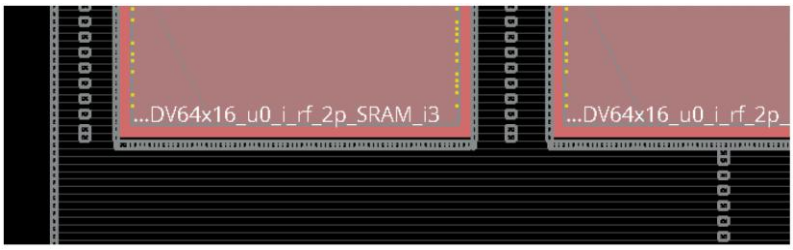
Add Endcap & Well Tap

1. Add endcap and well tap
   - innovus # > **source lab_script/add_endcaps.tcl**
   - Turn off the visibility of Route to observe the endcap





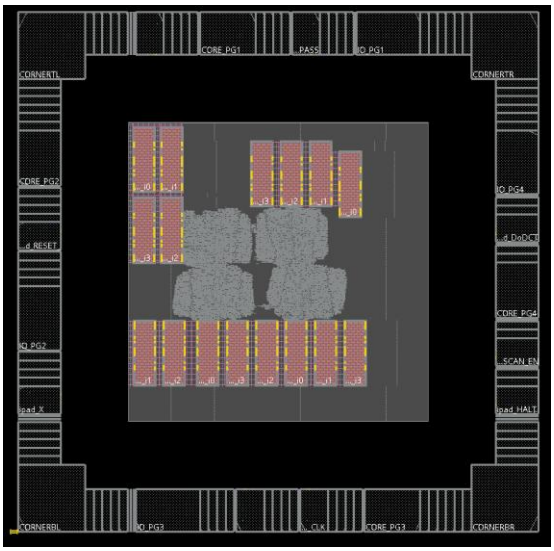
   - innovus # > **source lab_script/add_well_taps.tcl**



Placement

1. Source config.tcl and config_cts.tcl if you didn't source them before.
   - innovus # > **source lab_script/config.tcl**
   - innovus # > **source lab_script/config_cts.tcl**
2. Open ***Place → Place Standard Cell…***
   - Click OK button

Command line: **place_design** or **place_opt_design**



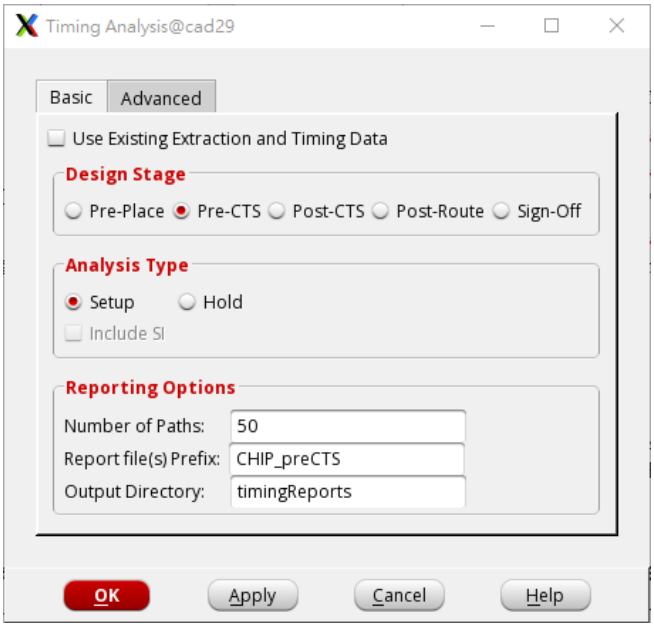
Change to **Physical view** to check if the cells are placed correctly.

3. In-Place Optimization – Before Clock Tree Synthesis

3.1 Open ***Timing → Report Timing…***

3.2 Perform First Encounter trial route to model the interconnection RC effects

- Design Stage ◆ pre-CTS
- Analysis Type ◆ Setup
- Click OK button



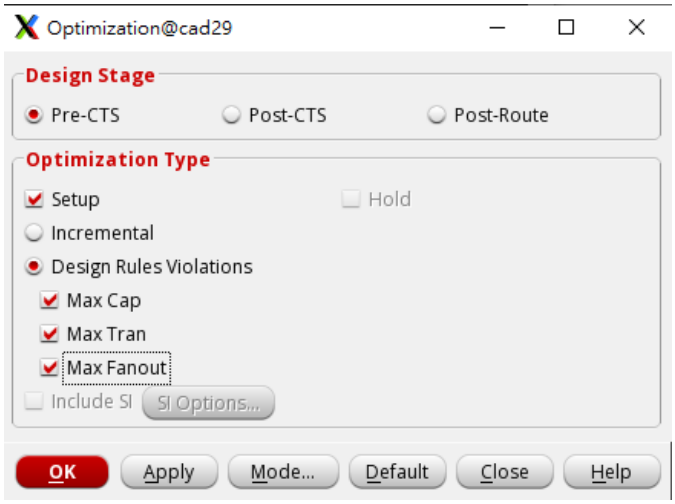
```
+--------------------+---------+---------+---------+---------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg |reg2cgate| in2reg  | reg2out | in2out  | default |
+--------------------+---------+---------+---------+---------+---------+---------+---------+
|          WNS (ns):| -1.025  | -1.025  |  1.444  |  1.762  |  1.616  |   N/A   |  0.000  |
|          TNS (ns):| -34.765 | -34.765 |  0.000  |  0.000  |  0.000  |   N/A   |  0.000  |
|    Violating Paths:|   48    |   48    |    0    |    0    |    0    |   N/A   |    0    |
|          All Paths:|  14785  |  14648  |   55    |   85    |    1    |   N/A   |    0    |
+--------------------+---------+---------+---------+---------+---------+---------+---------+

+----------------+------------------------------+------------------+
|                |             Real             |       Total      |
|    DRVs        +------------------+-----------+------------------|
|                |  Nr nets(terms)  | Worst Vio |  Nr nets(terms)  |
+----------------+------------------+-----------+------------------+
|   max_cap      |      7 (7)       |   -7.738  |     14 (14)      |
|   max_tran     |     80 (3081)    |   -1.293  |     80 (3081)    |
|   max_fanout   |      3 (3)       |   -6511   |      4 (4)       |
|   max_length   |      0 (0)       |     0     |      0 (0)       |
+----------------+------------------+-----------+------------------+
```

3.3 If the timing slack is negative, or there are DRVs, open ***ECO → Optimize Design…***

3.4 Perform pre-CTS IPO

- Design Stage ◆ pre-CTS
- Optimization Type
  - ◆ Setup
  - ◆ Design Rule Violations
  - ◆ Max Cap
  - ◆ Max Tran
  - ◆ Max Fanout
- Click OK button




```
+--------------------+---------+---------+---------+---------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg |reg2cgate| in2reg  | reg2out | in2out  | default |
+--------------------+---------+---------+---------+---------+---------+---------+---------+
|          WNS (ns):|  0.678  |  0.678  |  2.024  |  1.615  |  1.610  |   N/A   |  0.000  |
|          TNS (ns):|  0.000  |  0.000  |  0.000  |  0.000  |  0.000  |   N/A   |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |    0    |    0    |   N/A   |    0    |
|          All Paths:|  8253   |  8119   |   53    |   84    |    1    |   N/A   |    0    |
+--------------------+---------+---------+---------+---------+---------+---------+---------+

+----------------+------------------------------+------------------+
|                |             Real             |       Total      |
|    DRVs        +------------------+-----------+------------------|
|                |  Nr nets(terms)  | Worst Vio |  Nr nets(terms)  |
+----------------+------------------+-----------+------------------+
|   max_cap      |      0 (0)       |   0.000   |      0 (0)       |
|   max_tran     |      0 (0)       |   0.000   |      0 (0)       |
|   max_fanout   |      0 (0)       |     0     |      0 (0)       |
|   max_length   |      0 (0)       |     0     |      0 (0)       |
+----------------+------------------+-----------+------------------+
```

4. Save file
   - Open ***File → Save Design…***
   - Choose ◆ Innovus
   - File Name: **DBS/place.enc**

Clock Tree Synthesis (CTS)

1. Observe **CHIP_func.sdc** and **CHIP_scan.sdc** in the **design** directory
   1.1 Check **set_clock_latency** that estimated clock network delay is removed
   1.2 Check **set_ideal_network** is removed

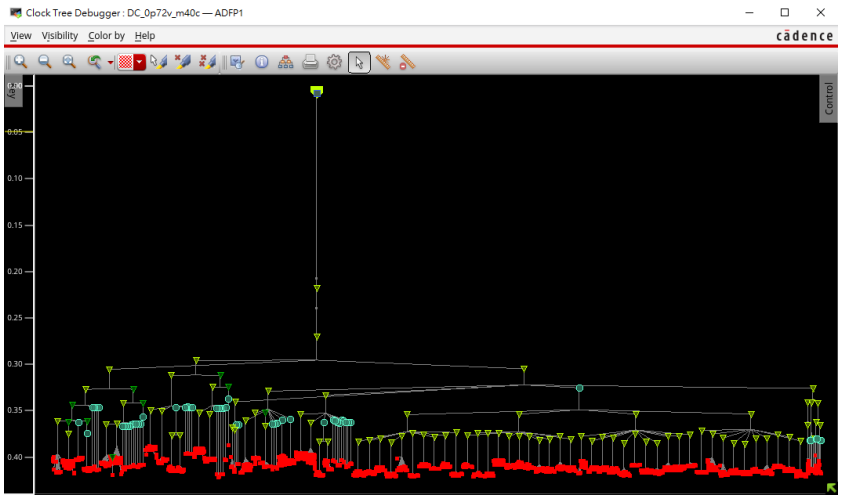
```
set sdc_version 1.2
#set_units -time ns -resistance kOhm -capacitance pF -voltage V -current uA
current_design CHIP
create_clock [get_ports {CLK}] -name CLK1 -period 100 -waveform {0 50}
create_clock  -name vclk -period 100 -waveform {0 50}
set_case_analysis 1 [get_ports {TEST_MODE}]
set_case_analysis 1 [get_ports {SCAN_EN}]
set_clock_uncertainty  -setup 0.5  [get_clocks {CLK1}]
set_clock_uncertainty  -hold 0.02  [get_clocks {CLK1}]
#set_clock_latency 0.3 [get_clocks {CLK1}]

set_input_delay     -max 3 -clock vclk [remove_from_collection [all_inputs] [get_ports CLK]]
set_input_delay     -min 0 -clock vclk [remove_from_collection [all_inputs] [get_ports CLK]]
set_output_delay    3 -clock vclk [all_outputs]

set_input_transition 0.1 [all_inputs]
set_drive 0.1 [all_inputs]
set_load -pin_load 20 [all_outputs]
set_max_fanout 8 [all_inputs]


#set_ideal_network [get_ports {CLK}]
```

2. Create spec file for CTS
   2.1 innovus #> **create_ccopt_clock_tree_spec -file ./ccopt.spec**
   2.2 innovus #> **source ./ccopt.spec**
   2.3 innovus #> **ccopt_design -cts**
3. CCOpt clock tree debugger
   - Open ***Clock → CCopt Clock Tree Debugger…***
   - Click OK button

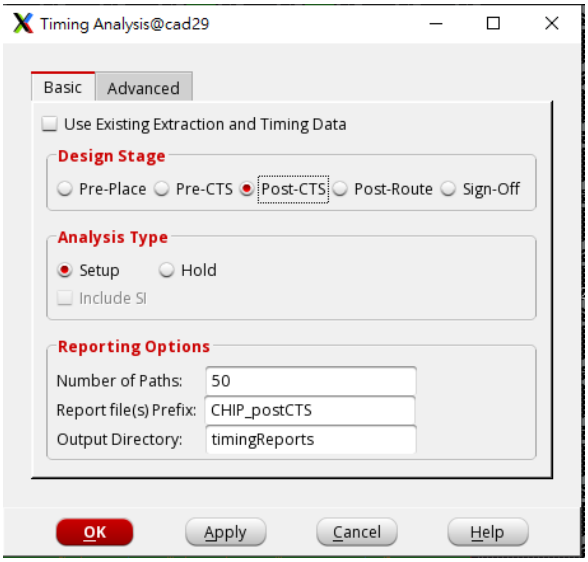


2. In-Place Optimization – After Clock Tree Synthesis

4.1 Open ***Timing → Report Timing…***

4.2 Perform First Encounter trial route to model the interconnection RC effects

- Design Stage ◆ post-CTS
- Analysis Type ◆ Setup
- Click OK button



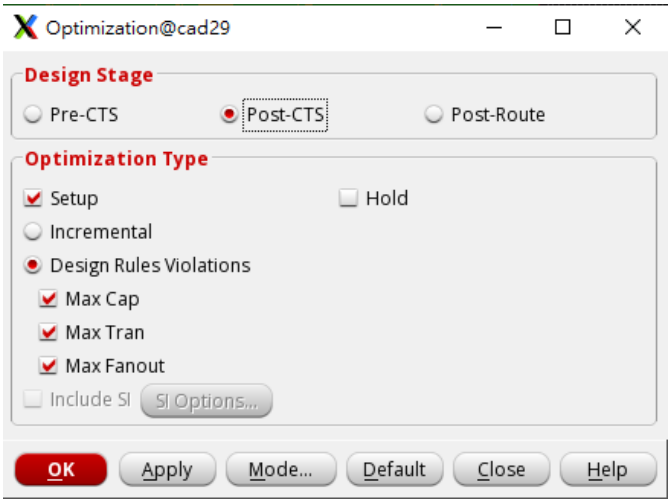
Setup mode	all	reg2reg	reg2cgate	in2reg	reg2out	in2out	default
WNS (ns):	0.752	0.752	2.073	1.874	1.148	N/A	0.000
TNS (ns):	0.000	0.000	0.000	0.000	0.000	N/A	0.000
Violating Paths:	0	0	0	0	0	N/A	0
All Paths:	8253	8119	53	84	1	N/A	0

DRVs	Real		Total	
	Nr nets(terms)	Worst Vio	Nr nets(terms)	
max_cap	0 (0)	0.000	0 (0)	
max_tran	0 (0)	0.000	0 (0)	
max_fanout	0 (0)	0	0 (0)	
max_length	0 (0)	0	0 (0)	

4.3 After CTS, further timing optimization is performed to meet timing constraints if there is negative timing slack or DRVs. Open ***ECO → Optimize Design…***

4.4 Perform post-CTS IPO

- Design Stage ◆ post-CTS
- Optimization Type
  - ◆ Setup
  - ◆ Design Rule Violations
  - ◆ Max Cap
  - ◆ Max Tran
  - ◆ Max Fanout
- Click OK button



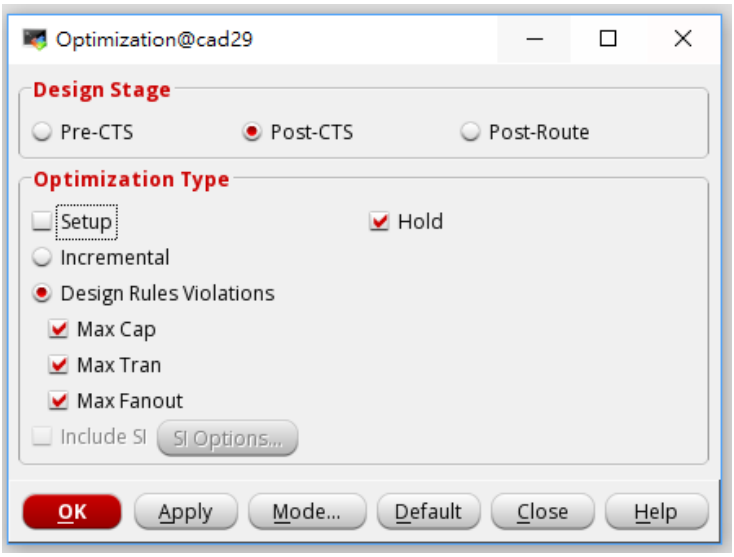
4.5 Verify if the hold time constraint is satisfied or not. Open ***Timing → Report Timing…***

- Design Stage ◆ post-CTS
- Analysis Type ◆ Hold
- Click OK button

4.6 If hold time slack is negative, open ***ECO → Optimize Design…***

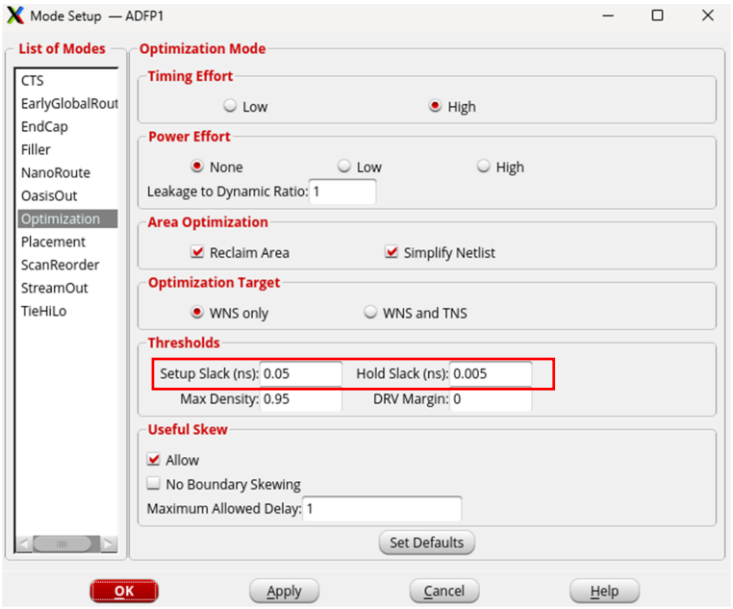
4.7 Perform post-CTS IPO

- Design Stage ◆ post-CTS
- Optimization Type
  - ◆ **Hold**
  - ◆ Design Rule Violations
  - ◆ Max Cap
  - ◆ Max Tran
  - ◆ Max Fanout
- Click OK button



If exist small negative slack (-0.000),

- Click Mode… button
- Thresholds
  - Setup slack: 0.05
  - Hold slack: 0.005



4.8 See timing reports in **timingReports** directory. For detail path report, see **CHIP_postCTS_reg2reg.tarpt.gz** (setup time check) and **CHIP_postCTS_reg2reg_hold.tarpt.gz** (hold time check).
DRV violations report files: ***.cap.gz** , ***.fanout.gz** , and ***.tran.gz**

3. Save file
   - Open ***File → Save Design…***
   - Choose ◆ Innovus
   - File Name: **DBS/cts.enc**

Add Tie Hi/Lo cell

1. Open ***Place → Tie HI/LO → Add…***
2. Cell Names already has two cells since it has been set in config.tcl. if it does not set in config.tcl, click Select button next to Cell Names and choose one **TIEHBWP20P90** and one **TIELBWP20P90**, click OK button.
3. Click OK button




```
INFO: Total Number of Tie Cells (TIEHBWP20P90) placed: 14
INFO: Total Number of Tie Cells (TIELBWP20P90) placed: 60
```

Routing

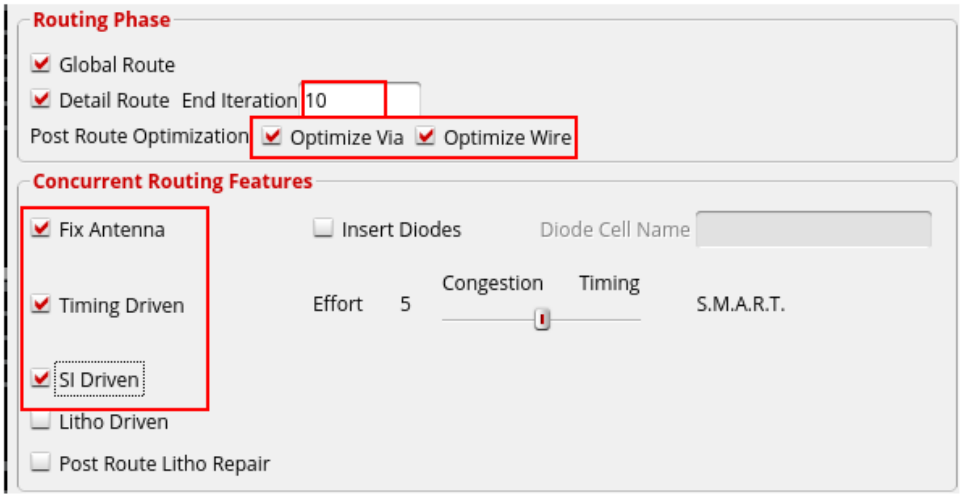
1. Open ***Route*** → ***NanoRoute*** → ***Route***
2. Nanoroute can prevent crosstalk effects and fix antenna rule violations, also it routes design to meet timing constraints

   2.1 Routing Phase
   - Detail Route End Iteration: **10**
   - Add ◆ **Optimize Via** and ◆ **Optimize Wire**

   2.2 Concurrent routing features
   - ◆ Fix Antenna
   - ◆ Timing Driven Effort: **5**
   - ◆ SI Driven

   2.3 Click OK button

If Innovus crash, cancel **Timing Driven**.



3. Save file
   - Open ***File*** → ***Save Design…***
   - Choose ◆ Innovus

- File Name: **DBS/route.enc**

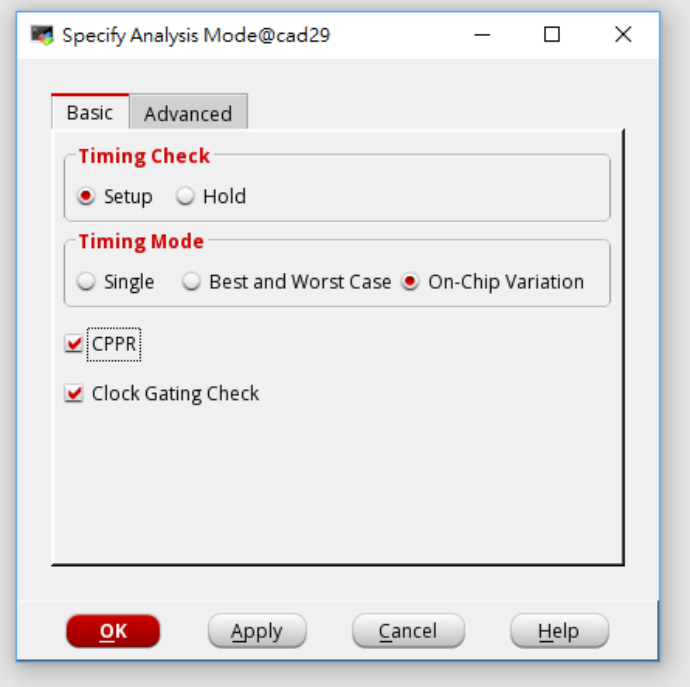
4. In-Place Optimization – After Detail Route

4.1 Set SI aware timing analysis

- innovus #> **set_db delaycal_enable_si true**

4.2 If did not set on-chip variation in config.tcl, open ***Tools*** → ***set Mode*** → ***Specify Analysis Mode…***

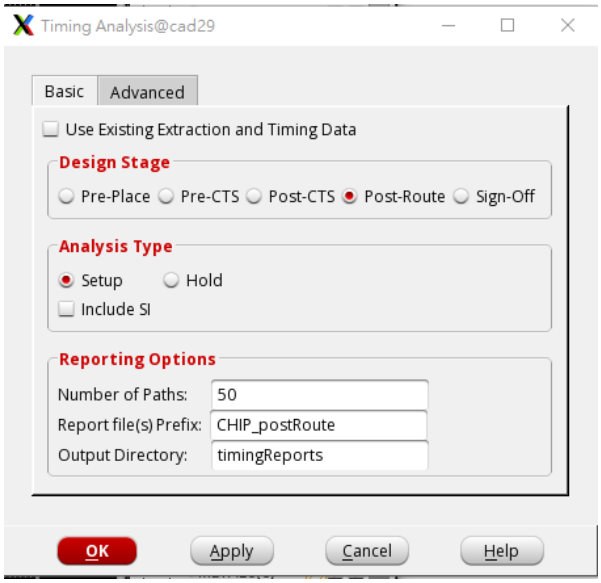
- Timing Mode ◆ On-Chip Variation
- ◆ CPPR
- Click OK button



4.3 Open ***Timing*** → ***Report Timing…***

4.4 Perform First Encounter trial route to model the interconnection RC effects

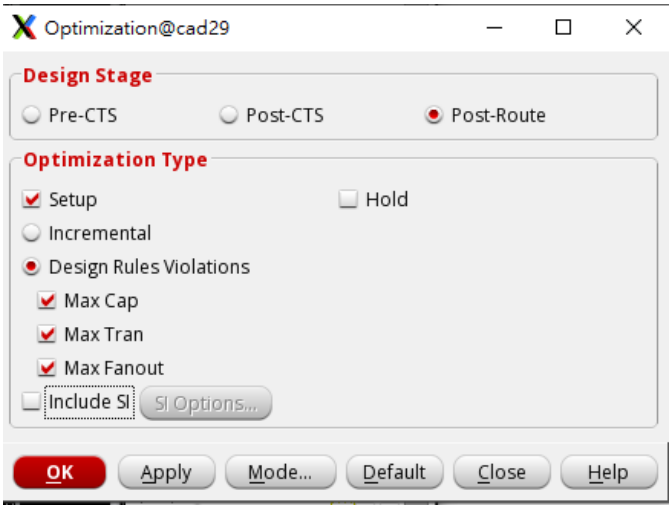
- Design Stage ◆ Post-Route
- Analysis Type ◆ Setup
- Click OK button



4.5 Further timing optimization is performed to meet timing constraints if there is negative timing slack or DRVs. Open ***ECO → Optimize Design…***

4.6 Perform post-Route IPO

- Design Stage ◆ post-Route
- Optimization Type
  - ◆ Setup
  - ◆ Design Rule Violations
  - ◆ Max Cap
  - ◆ Max Tran
  - ◆ Max Fanout
- Click OK button



4.7 Verify if the hold time constraint is satisfied or not. Open ***Timing → Report Timing…***

- Design Stage ◆ Post-Route
- Analysis Type ◆ Hold

- Click OK button

4.8 If hold time slack is negative, open ***ECO → Optimize Design…***

4.9 Perform post-Route IPO

- Design Stage ◆ post-Route
- Optimization Type
  - ◆ **Hold**
  - ◆ Design Rule Violations
  - ◆ Max Cap
  - ◆ Max Tran
  - ◆ Max Fanout
- Click OK button

5. Save file
   - Open ***File → Save Design…***
   - Choose ◆ Innovus
   - File Name: **DBS/route.enc**

Add Core Filler cells

1. Add core filler to improve electric effects of NWELL and PWELL
2. Open ***Place → Physical Cell → Add Filler…***
3. Cell Name(s) has been set in config.tcl. If it does not set in config.tcl, click Select button and select **FILL64BWP16P90 DCAP32BWP16P90 DCAP16BWP16P90 DCAP8BWP16P90 DCAP4BWP16P90 FILL2BWP16P90 FILL2BWP16P90LVT FILL1BWP16P90 FILL1BWP16P90LVT**.
   - Click OK button

Verify Geometry & Connectivity & Process Antenna

1. Verify geometry

   1.1 innovus # > **set_db check_drc_limit 10000**

   1.2 innovus # > **verify_drc**

   1.3 If there are many DRC errors (hundreds of DRC), perform full route again. Or directly type **route_opt_design** in the command line if using GUI doesn't work.

2. Verify connectivity

   2.1 Open ***Verify → Verify Connectivity …***

   2.2 Net Type ◆ **Regular Only**

   2.3 Nets ◆ **All**

   2.4 Unchosen ◆ **DanglingWire (Antenna)**

   2.5 Click Apply button

If there are dangling wires, use hot key T (shift + t) to fix it.

2.6 Net Type ◆ **Special Only**

2.7 Nets ◆ **All**

2.8 Unchosen ◆ **DanglingWire (Antenna)**

2.9 Click OK button

3. Verify process antenna

3.1 Open ***Verify → Verify Process Antenna…***

3.2 Click OK button

3.3 Just ignore the antenna violation since there is no antenna cell in ADFP

Add More Via

1. Add more VIA1 between M1 and M2 followpin
   - innovus # > **source lab_script/post_via_drop.tcl**
2. Check the DRC caused by adding VIA1
   - innovus # > **verify_drc**
3. Delete VIA1 that cause DRC
   - innovus # > **source lab_script/fix_mannual_via12.tcl**

Create Bump and Route

1. Already try to add bump before, add them now.
   - innovus # > **source lab_script/create_bump.tcl**
2. Verify geometry, connectivity and process antenna again.

Create Chip Boundary

1. Create chipBoundary layer due to chip height must be even integer of fin pitch in ADFP.
   - innovus # > **source lab_script/create_chipBoundary.tcl**
2. Save file
   - Open ***File → Save Design…***
   - Choose ◆ Innovus
   - File Name: **DBS/final.enc**

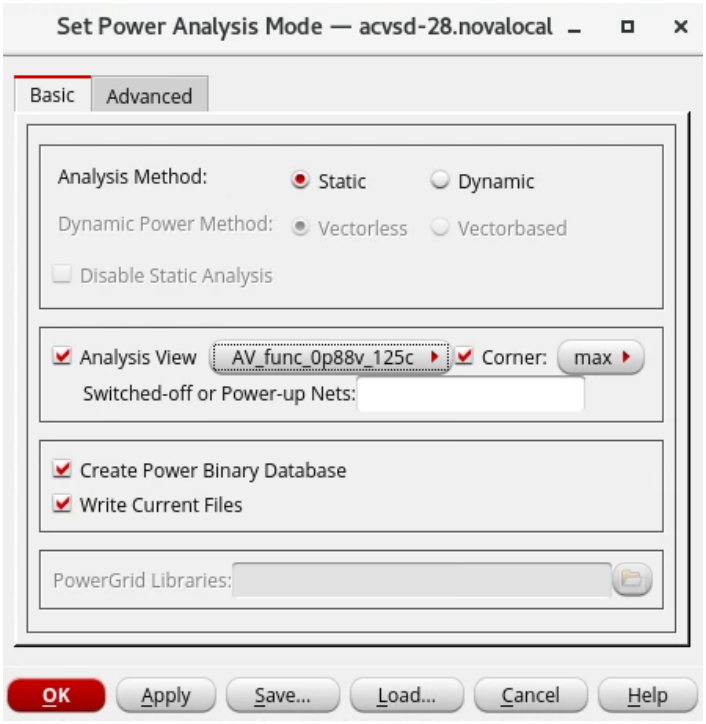
Output Data

1. Stream out sdf
   - innovus # > **source lab_script/write_sdf.tcl**
2. Stream out netlist
   - innovus # > **source lab_script/write_netlist.tcl**
   - **CHIP_pr.v** is used for post layout simulation. **CHIP_pg.v** is used for lvs.

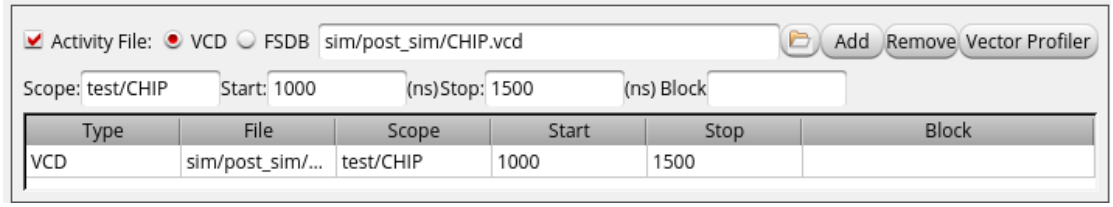
3. Stream out gds
   - innovus # > **source lab_script/write_stream.tcl**

Power Analysis

1. Open a new terminal and run post layout simulation to generate vcd file.
   - % **cd sim/post_sim**
   - % **sh run_postsim.sh**
2. Open ***Power → Power Analysis → Setup…***
   - Analysis Method: ◆ Static
   - Choose ◆ Analysis View **AV_func_0p88v_125c**
   - Choose ◆ Create Power Binary Database
   - Choose ◆ Write Current Files
   - Click OK button



3. Open ***Power → Power Analysis → Run…***
   - Choose ◆ Activity File: ◆ VCD
   - Set as following figure



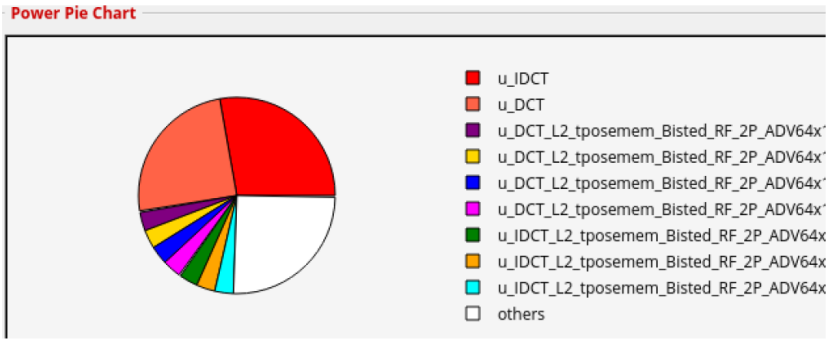
   - Click Add button
   - Click OK button

4. Observe the power consumption

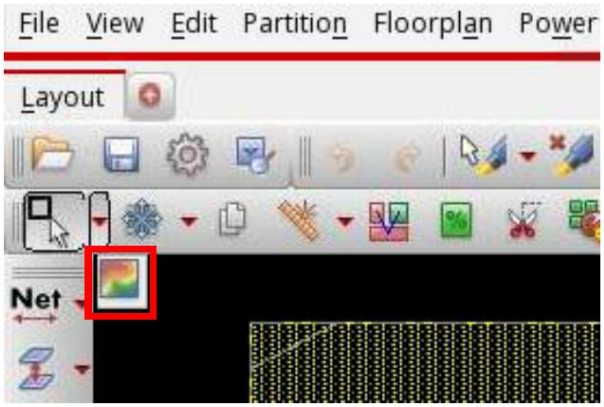
```
Total Power
-------------------------------------------------------------------
Total Internal Power:        9.54498097            62.9114%
Total Switching Power:       1.15808427             7.6330%
Total Leakage Power:         4.46903082            29.4556%
Total Power:                15.17209606
-------------------------------------------------------------------
```

5. Open ***Power → Report → Power Histogram…***

   ➢ Report power consumption ratio of each module in pie chart. Double click the pie chart to zoom in to lower hierarchy.



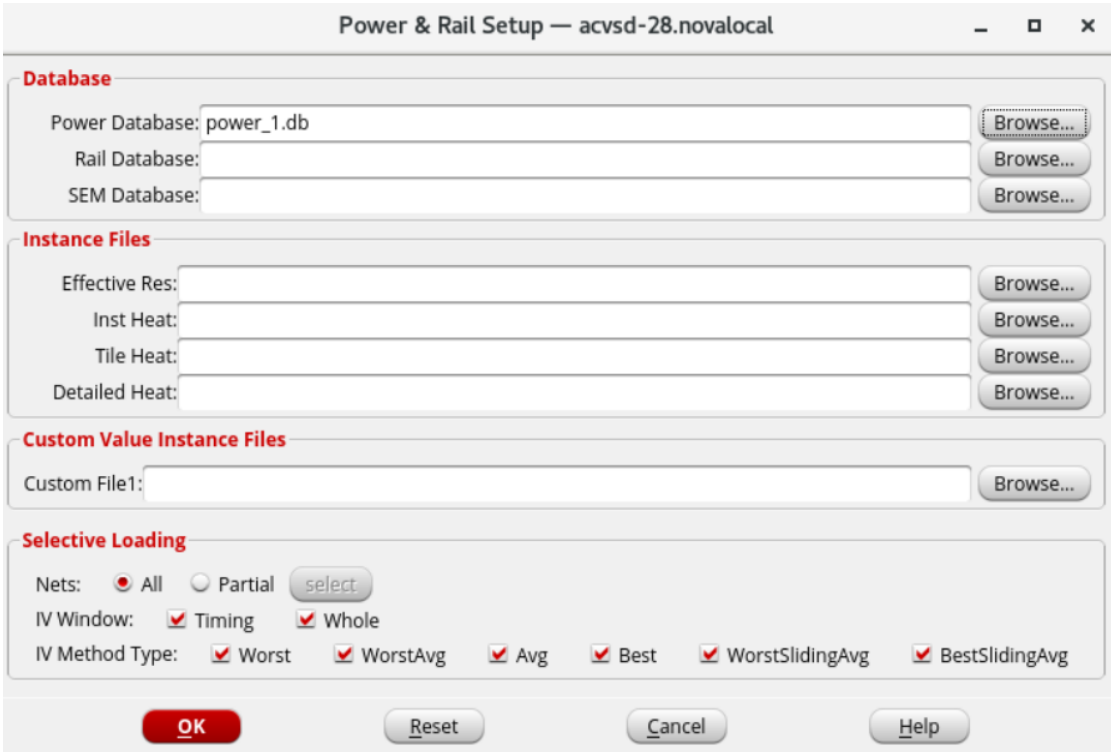
6. Open ***Power → Report → Power Rail Result…***



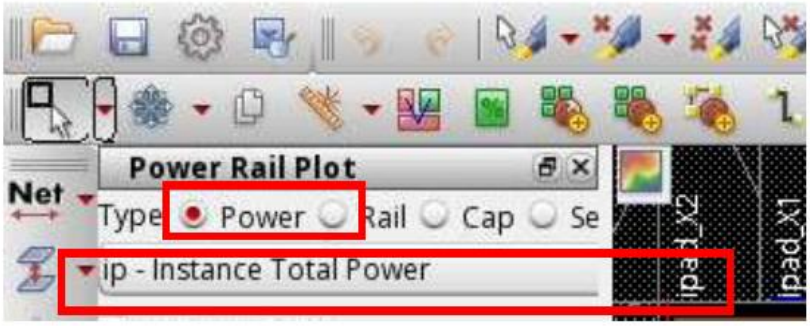
   ➢ Click  to open Power Rail Plot window, and click  to fix the window



- Click DB Setup button. It will show Power & Rail Setup window
- In the Power & Rail Setup window, Power Database: **power_1.db**
- Click OK button



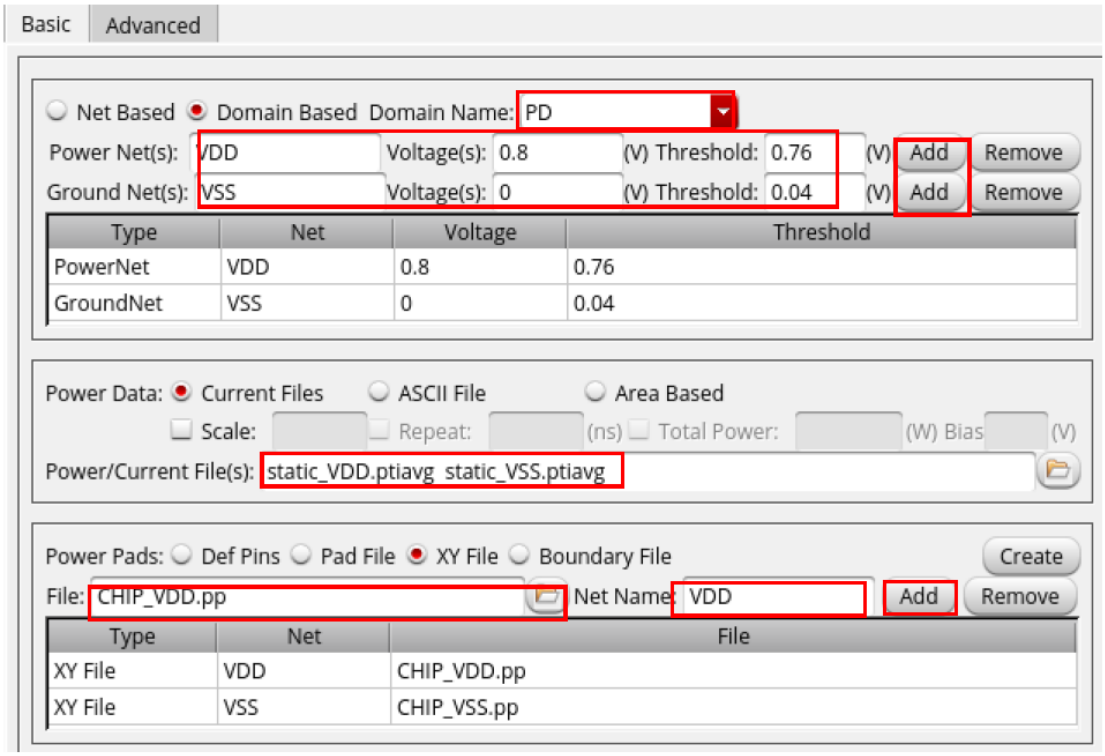
7. In the Power Rail Plot, Type: ◆ Power
   - Choose **ip - Instance Total Power, ip_s - Instance Switching Power, td - Transistion Density** and observe the layout. Change back to **none - Clear**



8. Open ***Power → Rail Analysis → Setup Rail Analysis…***
   - PowerGrid Libraries: **techonly.cl**
   - Click OK button



9. Open ***Power → Rail Analysis → Run Rail Analysis…***
   - ➢ Choose ◆ Domain Based Domain Name: **PD**
   - ➢ Power Net(s): **VDD** Voltage(s): **0.8** Threshold: **0.76** Click Add button
   - ➢ Ground Net(s): **VSS** Voltage(s): **0** Threshold: **0.04** Click Add button
   - ➢ Power/Current File(s): **static_VDD.ptiavg static_VSS.ptiavg**
   - ➢ Power Pads: ◆ XY File
   - ➢ File: **CHIP_VDD.pp** Net Name: **VDD** Click Add button
   - ➢ File: **CHIP_VSS.pp** Net Name: **VSS** Click Add button



   - ➢ Rail report of VDD is at PD_25C_avg_2/Reports/VDD/VDD.main.rpt. Rail report of VSS is at PD_25C_avg_2/Reports/VSS/VSS.main.rpt.

```
==============================
IR DROP ANALYSIS
==============================
Layer based IR Drop Report: See Report
Minimum, Average, Maximum IR drop: 0.796V 0.798V 0.800V
```

10. Click DB Setup button in the Power Rail Plot
    - ➢ Rail Database: **PD_25C_avg_2/VDD**
    - ➢ Click OK button



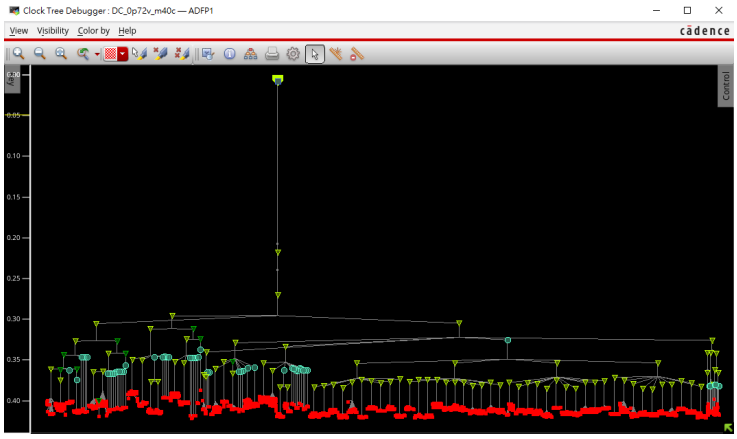
11. Back to Power Rail Plot, Type: ◆ Rail

- Choose **ir - IR Drop, rc - Resistor Current** and observe the layout. Change back to **none - Clear** and click  to close Power Rail Plot

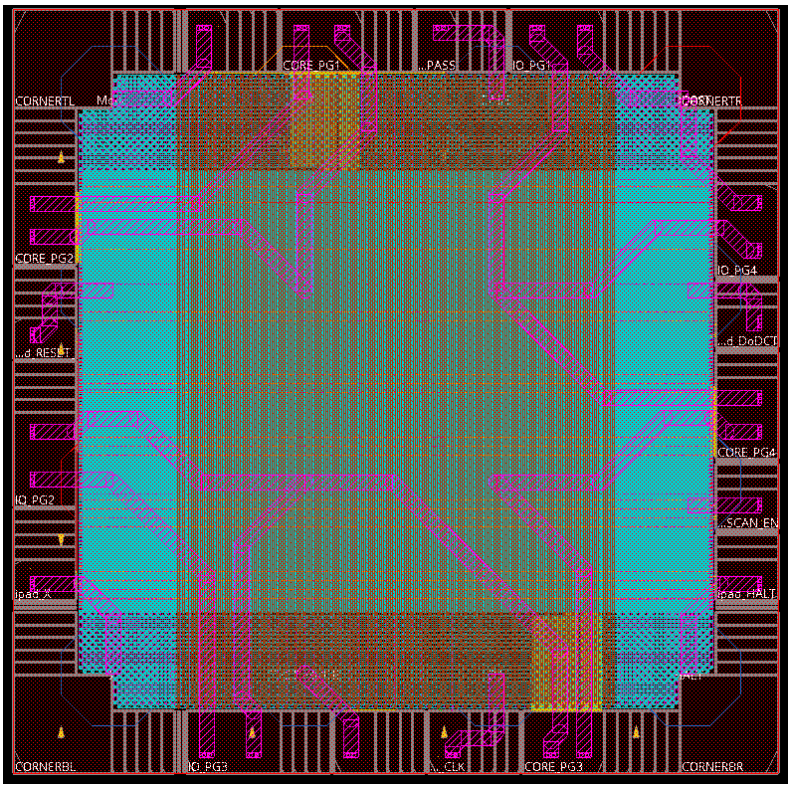


Checkpoints

1. Take a screenshot of the CCOpt clock tree debugger view.



2. Take a screenshot after routing (in physical view) and show post-route setup time and hold time analysis report (ensure the slack of setup time and the slack of hold time >= 0)



```
------------------------------------------------------------------
          timeDesign Summary
------------------------------------------------------------------

Setup views included:
 AV_func_0p72v_m40c AV_func_0p72v_125c

+--------------------+---------+---------+---------+---------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg |reg2cgate| in2reg  | reg2out | in2out  | default |
+--------------------+---------+---------+---------+---------+---------+---------+---------+
|           WNS (ns):|  0.754  |  0.754  |  2.073  |  1.858  |  1.207  |   N/A   |  0.000  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |  0.000  |  0.000  |   N/A   |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |    0    |    0    |   N/A   |    0    |
|         All Paths:|  8253   |  8119   |   53    |   84    |    1    |   N/A   |    0    |
+--------------------+---------+---------+---------+---------+---------+---------+---------+

+----------------+-------------------------------+------------------+
|                |             Real              |       Total      |
|    DRVs        +------------------+------------+------------------|
|                |  Nr nets(terms)  | Worst Vio  |  Nr nets(terms)  |
+----------------+------------------+------------+------------------+
|   max_cap      |      0 (0)       |   0.000    |      0 (0)       |
|   max_tran     |      0 (0)       |   0.000    |      0 (0)       |
|   max_fanout   |      0 (0)       |     0      |      0 (0)       |
|   max_length   |      0 (0)       |     0      |      0 (0)       |
+----------------+------------------+------------+------------------+

Density: 16.212%
       (100.000% with Fillers)
Total number of glitch violations: 0
------------------------------------------------------------------
```

```
------------------------------------------------------------------
          timeDesign Summary
------------------------------------------------------------------

Hold views included:
 AV_func_0p72v_m40c AV_func_0p72v_125c AV_func_0p88v_m40c
 AV_scan_0p88v_m40c AV_scan_0p88v_125c

+--------------------+---------+---------+---------+---------+---------+---------+---------+
|     Hold mode      |   all   | reg2reg |reg2cgate| in2reg  | reg2out | in2out  | default |
+--------------------+---------+---------+---------+---------+---------+---------+---------+
|           WNS (ns):|  0.000  |  0.000  |  0.066  |  0.003  |  3.983  |   N/A   |  0.000  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |  0.000  |  0.000  |   N/A   |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |    0    |    0    |   N/A   |    0    |
|         All Paths:|  14783  |  14648  |   53    |   85    |    1    |   N/A   |    0    |
+--------------------+---------+---------+---------+---------+---------+---------+---------+

Density: 16.212%
       (100.000% with Fillers)
------------------------------------------------------------------
```

3. Show verification results of DRC and connectivity.

```
 VERIFY DRC ..... Sub-Area: {56.544 452.352 84.816 480.624} 355 of 484  Thread : 25
 VERIFY DRC ..... Sub-Area: {508.896 84.816 537.168 113.088} 85 of 484  Thread : 22
 VERIFY DRC ..... Sub-Area: {56.544 84.816 84.816 113.088} 69 of 484  Thread : 9
 VERIFY DRC ..... Sub-Area: {113.088 508.896 141.360 537.168} 401 of 484  Thread : 2
 VERIFY DRC ..... Sub-Area: {537.168 480.624 565.440 508.896} 394 of 484  Thread : 31
 VERIFY DRC ..... Sub-Area: {84.816 452.352 113.088 480.624} 356 of 484  Thread : 25
 VERIFY DRC ..... Sub-Area: {56.544 508.896 84.816 537.168} 399 of 484  Thread : 12
 VERIFY DRC ..... Sub-Area: {508.896 537.168 537.168 565.440} 437 of 484  Thread : 18
 VERIFY DRC ..... Sub-Area: {480.624 480.624 508.896 508.896} 392 of 484  Thread : 30
 VERIFY DRC ..... Thread : 25 finished.
 VERIFY DRC ..... Thread : 24 finished.
 VERIFY DRC ..... Thread : 31 finished.
 VERIFY DRC ..... Sub-Area: {84.816 508.896 113.088 537.168} 400 of 484  Thread : 12
 VERIFY DRC ..... Thread : 12 finished.
 VERIFY DRC ..... Sub-Area: {508.896 508.896 537.168 537.168} 415 of 484  Thread : 30
 VERIFY DRC ..... Thread : 30 finished.

  Verification Complete : 0 Viols.
```

```
******** Start: VERIFY CONNECTIVITY ********
Start Time: Sat Nov 23 09:58:12 2024

Design Name: CHIP
Database Units: 2000
Design Boundary: (0.0000, 0.0000) (614.4300, 613.6800)
Error Limit = 1000; Warning Limit = 50
Check all nets
Use 32 pthreads
*** 09:58:12 *** Building data for Net VDD
*** 09:58:12 *** Building data for Net VSS

Begin Summary
  Found no problems or warnings.
End Summary

End Time: Sat Nov 23 09:58:14 2024
Time Elapsed: 0:00:02.0

******** End: VERIFY CONNECTIVITY ********
  Verification Complete : 0 Viols.  0 Wrngs.
  (CPU Time: 0:00:04.3  MEM: 0.000M)
```

4. **Due Tuesday, Apr. 28, 23:59**
5. Submit to NTU COOL in pdf format.